

MULTITRACK ECM

FOR
DUMMIES®

**LIAM
KIRKPATRICK**

A Reference for the Rest of Us!



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Operating Guide (Basic)

This section provides the basic information needed to run the ECM system. Note that it does NOT provide the information needed to fix issues that come up along the way – for that, see the “building guide” later on in this document.

Arriving to ICF:

Cable setup

This is in the case where the ECM is not already plugged in and set up in the freezer.

Cables bold must be passed through freezer wall. It's easiest to do this as a single bundle, passing from inside the freezer to the outside.

The following items need to be connected to the computer:

- USB cable to LCR
- USB cable to SMU
- USB cable to MCC DAQ
- Wifi antenna (special two screw in points)
- USB keyboard/mouse dongle (for warm area controls)
- DP to VGA cable for warm area monitor
- **USB to Zaber cable**
- **HDMI cable for freezer monitor**
- **USB extenders (x3) for interior keyboard, mouse, gaming controller**

Plug into the MCCDAQ:

- **2-wire bundle for laser control**
- **6-wire bundle for encoder control**
- **3-wire bundle for button reading**

Plug into the SMU:

- **4 measurement Cables**

Plug into the LCR:

- **2 measurement cables**

Plug into power (inside freezer):

Note that all interior power runs through the Uninterruptable Power Supply (UPS) stored under the ECM table in a cooler.

- 2x old-school zaber power cables
- 1x new-school zaber power cable

- 1x zaber x-controller power cable

Plug into power (outside freezer):

Also all plugged into UPS, outside of freezer

- Computer
- Monitor
- LCR
- SMU

Other Setup

Outside computer station:

- Remove yellow panels covering box so cables can be attached.
- Set up monitor (remove from box)
- Find Bluetooth keyboard and mouse. Turn these on! Ensure the Bluetooth USB dongle is plugged in

Arriving to ICF – how to start:

This is to be completed after the steps above, or if the cables and overall setup is already in place.

- Remove Plastic cover from ECM.
- Turn on the UPS systems both inside and outside of the freezer. Inside the freezer, this will mean getting under the ECM table to plug the UPS that's inside of the cooler in to the wall (can be tricky with stiff cord).
- Plug in the TV (also tricky with stiff cord).
- Power up Computer, LCR, SMU, outside monitor.
- Power up inside-freezer monitor
- Launch program and you're off!

Launching the program

Boot up computer:

- Password is “ecm2023”

To launch the ECM GUI, please follow the following steps:

- Launch Anaconda Powershell
 - (alternatively, you can launch a standard terminal window and type “miniconda/Scripts/activate base” – does the same thing)
- Open the correct Python Environment with “conda activate ecm”
- Use “cd” commands to navigate to folder with files
- To run the GUI, it’s as simple as typing “Python ECMGUI_vXX.py”
- To edit the GUI, use Spyder. Open by typing “Spyder” in a second anaconda PowerShell window.
 - NOTE: I advise against running the gui from within spyder – it seems to glitch more when run this way.
 - **MAKE SURE YOU ARE EDITING THE RIGHT VERSION OF THE CONTROL SCRIPT**

Turn on screen inside freezer

- Do this AFTER booting up the outside screen. Otherwise the computer pixel dimensions get weird until you reboot.
- Press the power button (bottom edge, bottom left of TV) once to turn on the TV. Give it a couple of seconds to turn on, then press the same button a second time to switch the input to HDMI1
- Screen should be working and connected. First debugging check is to ensure the HDMI cord is plugged into both the TV and Computer.

Using the GUI

End of day Checklist

At the end of the day:

- Ensure no motors are moving! I do this by running the script one last time from outside the freezer. If the motors all home without error, you know none of the manual controls are accidentally on.
- Copy data to hard drive. There should be spare USB ports on the back of the computer to plug into. I tend to just copy the whole outputs data file, but just the day is also fine.
- Turn off LCR and SMU (power buttons on front side of yellow box (closer to the door))
- Turn off computer (normal windows shut down).
- Leave all power on. The TV turns itself off automatically, but keeping everything warm with power is worth it.

Short-term Takedown Instructions

Leaving ICF, don't want to pull cables because ECM will be used again soon.

- Complete end-of-day checklist above.
- Turn off both UPS systems (hold power button) and unplug them. Lights should go off.
- Unplug monitor inside.
- Place plastic cover over ECM inside of freezer, protecting from drips.

Full Takedown Instructions

Leaving ICF for a longer amount of time (>1 month?). This is a full takedown that clears space outside of freezer and removes cables from walls.

Really, this involves reversing the set-up instructions. Follow the check list below.

(coming soon – Emma will have to you next week if needed)

Building Guide (Advanced)

Python

We are using python 3 for this system. Note that this was installed with miniconda, not anaconda. I forget why, but this simpler install fixed some issue at some point... You'll have to just take my word for it. All of the system is contained within the "ecm" conda environment.

ECMGUI

This is our master code that controls the entire system. The scrips are laid out as follows:

MCC DAQ

Encoder

Laser

Button

SMU

LCR

Gaming Controller

Zaber Motors

Uninterruptable Power Supply (UPS)